Gebze Technical University

Department of Mechanical Engineering

Recitation 1

Introduction and Fundamentals

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Self-study guide. This recitation contains 25 questions with worked solutions. Read each question and follow the explanation directly below it. The solutions develop the physical ideas, modeling assumptions and calculations used in Week 1.

Question 1 | Historical milestones

Select the most appropriate answer from the multiple choices given.

(a) The first experiments on simple pendulums were conducted by
(a) Galileo (b) Pythagoras (c) Aristotle.

(b) The *Philosophiae Naturalis Principia Mathematica* was published by
(a) Galileo (b) Pythagoras (c) Newton.

(c) Mode shapes of plates, by placing sand on vibrating plates, were first observed by
(a) Chladni (b) D'Alembert (c) Galileo.

Solution 1

(a) **Galileo, choice (a).** His pendulum experiments helped establish the study of oscillatory motion. For small swings, a pendulum's period is approximately independent of amplitude; this is an approximation, not an exact statement for large swings.

(b) **Newton, choice (c).** The *Principia* presented the laws of motion. In vibration modeling, these laws connect forces and moments to translational and angular acceleration.

(c) **Chladni, choice (a).** Sand on a vibrating plate collects near nodal lines, where the motion is small. The resulting patterns make the plate's vibration mode shapes visible.

Question 2 | Useful and harmful vibration

Give two examples each of the bad and the good effects of vibration.

Solution 2

Harmful effects

(a) Repeated vibration creates cyclic stresses in machine parts. These stresses can cause fatigue cracks and eventual failure.

(b) Vibration of a vehicle or machine can cause human discomfort and reduce the accuracy of measurements or machining.

Useful effects

(a) Vibrating screens separate particles: repeated motion helps smaller particles pass through the openings.

(b) Musical instruments use vibrating strings, membranes or air columns to produce sound.

The same physical phenomenon can be useful or harmful depending on the intended function. Other appropriate examples are also valid.

Question 3 | The three elementary parts

What are the three elementary parts of a vibrating system?

Solution 3

(a) **Mass or inertia element:** resists changes in motion and stores kinetic energy. For translation, $T = \frac{1}{2}m\dot{x}^2$; for rotation, $T = \frac{1}{2}J\dot{\theta}^2$.

(b) **Spring or elastic element:** provides a restoring force and stores potential energy. A linear spring stores $U = \frac{1}{2}kx^2$, where x is its deformation.

(c) **Damper or dissipative element:** removes mechanical energy, usually converting it to heat. For a viscous damper, the energy-loss rate is $c\dot{x}_{\text{rel}}^2$.

During vibration, energy is exchanged between inertia and elastic elements. Damping causes the total mechanical energy to decrease when no external work is supplied. A physical component can contribute more than one of these effects.

Question 4 | Degrees of freedom

Define the number of degrees of freedom of a vibrating system.

Solution 4

The number of degrees of freedom (DOF) is the **minimum number of independent coordinates** needed to describe the configuration of all parts of a system.

1. Identify the possible motion

List the translations and rotations that may occur.

2. Apply the constraints

Fixed supports, guides and rigid links may make some coordinates dependent on others. Dependent coordinates do not add DOF.

3. Count the remaining independent coordinates

A simple pendulum constrained to swing in a plane needs only its angle θ: the fixed length determines the bob's position once θ is known. It therefore has one DOF, even though the bob moves both horizontally and vertically.

Question 5 | Discrete and continuous systems

What is the difference between a discrete and a continuous system? Is it possible to solve any vibration problem as a discrete one?

Solution 5

Discrete model: motion is described by a finite set of coordinates, such as $x_1(t), x_2(t), \ldots, x_n(t)$. The model has finitely many DOF.

Continuous model: motion varies throughout the body and is described by a field, such as beam deflection $w(s,t)$ along position s. Ideally, it has infinitely many DOF.

Can a discrete model be used? A continuous system can often be approximated by a finite model, for example by dividing it into smaller elements. The finite model is generally an approximation, not an exact representation of every possible motion.

Refine the model and check whether the response of interest changes appreciably. Adequate accuracy depends on the purpose of the analysis and the frequency range being studied.

Question 6 | Free and forced vibration

Fill in the blank with the proper word.

(a) If a system vibrates due to initial disturbance only, it is called _____ vibration.

(b) If a system vibrates due to an external excitation, it is called _____ vibration.

Solution 6

(a) **Free vibration.** An initial displacement or velocity supplies the starting energy. After release, no continuing external excitation acts. The motion is governed by the system's inertia, elasticity and damping.

(b) **Forced vibration.** An external excitation acts during the response. This may be an applied force or prescribed support motion; it need not be sinusoidal.

Separate the classifications. Free vibration may be damped or undamped. "Free" describes the excitation; "damped" describes energy dissipation. A damped system released from an initial displacement still undergoes free vibration.

Question 7 | When damping matters

In vibration analysis, can damping always be disregarded?

Solution 7

No. Damping removes mechanical energy and can change the response substantially.

1. Consider free vibration

Without damping, an ideal vibrating system retains its mechanical energy. With damping, its free motion loses energy and usually decays. Ignoring damping cannot predict this decay.

2. Consider forced vibration

When the excitation is near a natural frequency, damping strongly influences the response amplitude. Neglecting it can give a poor estimate of the motion.

3. Match the assumption to the purpose

An undamped approximation may help explain energy exchange or provide an initial estimate. It is acceptable only when the neglected energy loss has little effect on the quantity and time interval of interest.

Question 8 | Recognizing nonlinearity

Can a nonlinear vibration problem be identified by looking at its governing differential equation?

Solution 8

Yes, when the governing equation is available. Inspect how the unknown motion and its derivatives occur.

1. Recognize a linear equation

An equation such as

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

is linear in x: the unknown and its derivatives appear to the first power and are not multiplied together. Coefficients that depend only on time do not by themselves make an equation nonlinear.

2. Look for nonlinear terms

Terms such as x^3, $x\dot{x}$ or $\sin\theta$ are nonlinear. The exact pendulum restoring term $mgl\sin\theta$ is an example.

3. Distinguish a model from its approximation

Replacing $\sin\theta$ by θ gives a linear approximation for small angles. The original physical model remains nonlinear. Superposition applies to linear equations; it does not generally apply to nonlinear ones.

Question 9 | Deterministic and random vibration

What is the difference between deterministic and random vibration? Give two practical examples of each.

Solution 9

Deterministic vibration: the excitation can be specified as a known function of time. With the model and initial conditions known, the response can be calculated as a function of time.

- Periodic forcing from a machine operating at a prescribed constant speed.
- Response to a known impact pulse with a specified force history.

Random vibration: the excitation history is not known in advance in full detail. It is described using statistical quantities.

- Vehicle excitation from road roughness represented statistically.
- Structural excitation from turbulent wind represented statistically.

Deterministic does not mean periodic: a known single pulse is deterministic but does not repeat.

Question 10 | Stiffness and damping constant

Define spring stiffness and damping constant.

Solution 10

1. Spring stiffness

For a linear spring, force magnitude is proportional to deformation:

$$F = k\delta \quad \Longrightarrow \quad \boxed{k = F/\delta}, \qquad [k] = \mathrm{N/m}.$$

Use the deformation *of the spring*, not an arbitrary position coordinate. For a nonlinear force law, the local tangent stiffness at equilibrium δ_0 is

$$k_t = \left.\frac{dF}{d\delta}\right|_{\delta_0}.$$

2. Viscous damping constant

For a linear viscous damper, force magnitude is proportional to the relative speed of its ends:

$$F = c\,v_{\mathrm{rel}} \quad \Longrightarrow \quad \boxed{c = F/v_{\mathrm{rel}}}, \qquad [c] = \mathrm{N\,s/m}.$$

In a signed force equation, the force opposes relative motion. If both ends move together at the same velocity, the damper produces no viscous force.

Question 11 | Common damping mechanisms

What are the common types of damping?

Solution 11

(a) **Viscous damping.** In the linear model, resisting force is proportional to relative velocity. For one endpoint,

$$F_d = -c\,v_{\mathrm{rel}}.$$

Fluid films and dashpots can often be represented by this model under suitable assumptions.

(b) **Coulomb or dry-friction damping.** During sliding, friction has approximately constant magnitude $\mu_f N$ and acts opposite the sliding velocity. Its direction reverses when sliding reverses. At rest, the friction law must allow sticking.

(c) **Material or hysteretic damping.** Internal losses occur during repeated deformation. The loading and unloading paths differ; the area enclosed by a force–displacement loop equals the mechanical energy lost in one cycle.

All three dissipate energy, but they obey different force laws. One constant viscous coefficient does not describe every damping mechanism exactly.

Question 12 | modeling the human body

A study of the response of a human body subjected to vibration/shock is important in many applications. In a standing posture, the masses of head, upper torso, hips, and legs and the elasticity/damping of neck, spinal column, abdomen, and legs influence the response characteristics. Develop a sequence of three improved approximations for modeling the human body.

Solution 12

This modeling question admits more than one justified answer. The following sequence studies **small vertical motion** about standing equilibrium. Each compliant connection is represented by a spring and a damper in parallel.

1. One-DOF approximation

Combine the body segments into one mass m_b. Connect it to the floor through an effective support stiffness k_b and damping c_b. Its vertical displacement x_b is the only coordinate. Relative motion between body segments is suppressed.

2. Two-DOF approximation

Separate the upper body (head and torso) from the lower body (hips and legs). Use masses m_u, m_l and independent coordinates x_u, x_l.

Connect the upper mass to the lower mass through effective spinal/abdominal stiffness and damping. Connect the lower mass to the floor through effective leg/support stiffness and damping. The internal connection deforms by $x_u - x_l$; its relative velocity is $\dot{x}_u - \dot{x}_l$.

3. Four-DOF approximation

Retain four masses: head, torso, hips and legs. Use x_h, x_t, x_p, x_l as their vertical coordinates. The model sequence is

$$\text{head} \longleftrightarrow \text{torso} \longleftrightarrow \text{hips} \longleftrightarrow \text{legs} \longleftrightarrow \text{floor}.$$

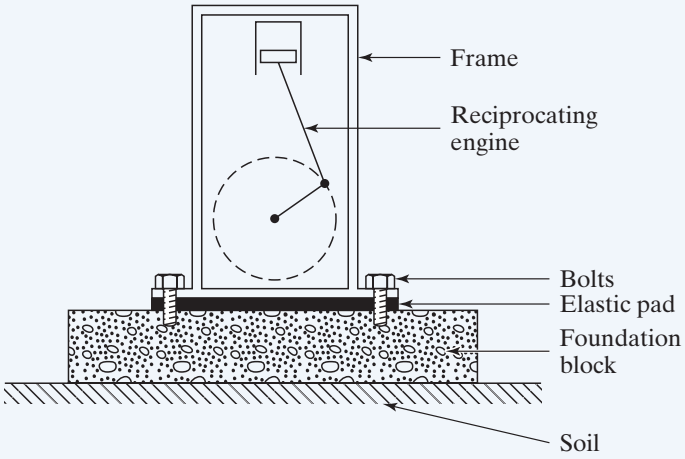
Each arrow represents an effective spring and damper connection. The neck connects head to torso; trunk and leg compliance connect the remaining segments. An effective foot/support connection joins the legs to the floor.

4. Interpret the refinement

The models have 1, 2 and 4 DOF. Each refinement permits additional relative motion. If the floor moves by a prescribed $y(t)$, the lowest connection deforms relative to $y(t)$; prescribed input motion is not an extra unknown DOF. Numerical properties require additional data.

Question 13 | An engine and its foundation

A reciprocating engine is mounted on a foundation as shown below. The unbalanced forces and moments developed in the engine are transmitted to the frame and the foundation. An elastic pad is placed between the engine and the foundation block to reduce the transmission of vibration. Develop two mathematical models of the system using a gradual refinement of the modeling process.



A reciprocating engine on a foundation.

Solution 13

1. Choose the motion and coordinates

One valid approach starts with vertical translation. Assume small motion, a rigid engine/frame, and a linear pad. Measure all displacements from static equilibrium, so the balanced gravity terms do not appear in the incremental equations.

2. Model 1: fixed foundation, one DOF

Let m_e be the engine/frame mass and $x_e(t)$ its vertical displacement. Represent the pad by k_p, c_p in parallel and initially treat the foundation as fixed.

With upward positive, the pad forces on the engine are $-k_p x_e$ and $-c_p \dot{x}_e$. Applying Newton's second law gives

$$m_e \ddot{x}_e = F(t) - c_p \dot{x}_e - k_p x_e,$$

or

$$\boxed{m_e \ddot{x}_e + c_p \dot{x}_e + k_p x_e = F(t).}$$

This model describes engine motion relative to a fixed foundation. The pad force transmitted to the foundation is $k_p x_e + c_p \dot{x}_e$.

Solution 13 (continued)

3. Model 2: movable foundation, two DOF

Allow the foundation mass m_f to move with an independent coordinate $x_f(t)$. Keep the pad between engine and foundation. Model the soil by a spring k_s and damper c_s connecting the foundation to fixed ground.

The connections are therefore: engine/frame mass — pad — foundation mass — soil — ground. The pad deformation and relative velocity are

$$\delta_p = x_e - x_f, \qquad \dot{\delta}_p = \dot{x}_e - \dot{x}_f.$$

4. Write the engine force balance

The pad force opposes the engine's motion relative to the foundation:

$$\boxed{m_e\ddot{x}_e + c_p(\dot{x}_e - \dot{x}_f) + k_p(x_e - x_f) = F(t).}$$

5. Write the foundation force balance

The force exerted by the pad on the foundation is opposite to its force on the engine. The soil forces oppose foundation motion:

$$m_f\ddot{x}_f = c_p(\dot{x}_e - \dot{x}_f) + k_p(x_e - x_f) - c_s\dot{x}_f - k_s x_f.$$

Collecting terms,

$$\boxed{m_f\ddot{x}_f + c_p(\dot{x}_f - \dot{x}_e) + k_p(x_f - x_e) + c_s\dot{x}_f + k_s x_f = 0.}$$

6. Explain what the refinement adds

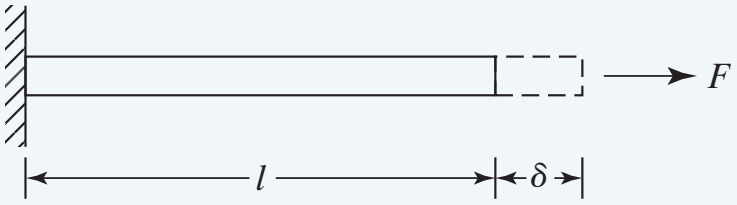
The second model captures foundation translation and soil flexibility. The two equations are coupled because the pad depends on *relative* motion. Setting $x_f = 0$ in the engine equation recovers Model 1.

These are translational models. They omit the rocking caused by the unbalanced moments mentioned in the question. To study that response, add angular coordinates and rotational inertias for the engine/frame and foundation, together with the moment effects of their supports. The chosen coordinates must match the response being studied; no single model is unique.

The task here is to formulate and interpret the models. A complete vibration analysis then solves the governing equations using the excitation and initial conditions, and interprets the predicted motion or transmitted force. Check units, limiting cases and agreement with measurements before deciding whether the model needs further refinement.

Question 14 | Axial stiffness of a rod

Find the equivalent spring constant of a uniform rod of length l, cross-sectional area A, and Young's modulus E subjected to an axial tensile (or compressive) force F.



Axial force and end deformation of a uniform rod.

Solution 14

Assume a uniform, linearly elastic rod with small axial deformation. Use F and δ as the magnitudes of axial force and end deformation.

1. Find the axial stress

Uniform stress is force divided by cross-sectional area:

$$\sigma = \frac{F}{A}.$$

2. Relate stress to strain

Hooke's law gives $\sigma = E\varepsilon$, hence

$$\varepsilon = \frac{\sigma}{E} = \frac{F}{AE}.$$

3. Find the end deformation

Strain is deformation per unit length: $\varepsilon = \delta/l$. Therefore,

$$\delta = \varepsilon l = \frac{Fl}{AE}.$$

4. Identify the stiffness and check its units

Compare with $F = k\delta$:

$$\boxed{k = \frac{F}{\delta} = \frac{AE}{l}}, \qquad \frac{\mathrm{m}^2(\mathrm{N/m}^2)}{\mathrm{m}} = \mathrm{N/m}.$$

A larger area or modulus increases stiffness; a longer rod is more flexible. Under compression, this axial model applies while the straight configuration is maintained; it does not describe buckling.

Question 15 | Translational and torsional quantities

State the parameters corresponding to m, c, k, and x for a torsional system.

Solution 15

1. Identify the angular coordinate and inertia

Translation x is replaced by angular displacement θ (rad). Mass m is replaced by the mass moment of inertia J_O about the fixed rotation axis ($\mathrm{kg\,m^2}$):

$$\sum F = m\ddot{x} \quad \longleftrightarrow \quad \sum M_O = J_O\ddot{\theta}.$$

2. Identify rotational damping and stiffness

The translational coefficient c is replaced by c_θ ($\mathrm{N\,m\,s/rad}$); the stiffness k is replaced by k_θ ($\mathrm{N\,m/rad}$). With a stationary support,

$$M_d = -c_\theta\dot{\theta}, \qquad M_s = -k_\theta\theta.$$

The corresponding equations are

$$m\ddot{x} + c\dot{x} + kx = F(t), \qquad J_O\ddot{\theta} + c_\theta\dot{\theta} + k_\theta\theta = M(t).$$

3. Connect torsional stiffness to a circular shaft

For a uniform, linearly elastic circular shaft of length L, shear modulus G and polar *area* moment J_p, a holding torque M produces twist

$$\theta = \frac{ML}{GJ_p}.$$

Comparing $M = k_\theta\theta$ gives

$$\boxed{k_\theta = \frac{GJ_p}{L}}.$$

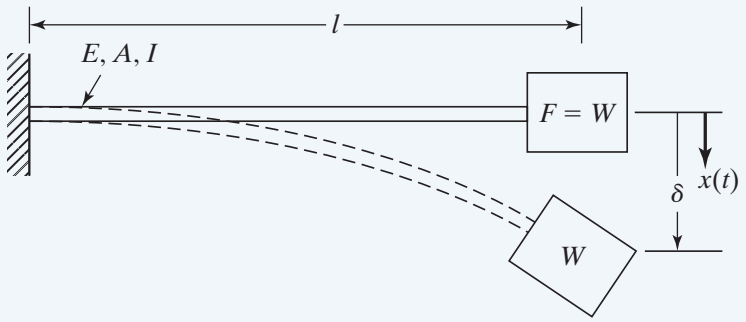
For a solid circular section, $J_p = \pi d^4/32$; for a hollow section, $J_p = \pi(d_o^4 - d_i^4)/32$. These are area quantities in $\mathrm{m^4}$.

4. Distinguish stiffness geometry from mass inertia

The mass moment $J_O = \int r^2\,dm$ has units $\mathrm{kg\,m^2}$, whereas $J_p = \int r^2\,dA$ has units $\mathrm{m^4}$. They are not interchangeable. The kinetic energy is $T = \frac{1}{2}J_O\dot{\theta}^2$; the elastic energy is $U = \frac{1}{2}k_\theta\theta^2$.

Question 16 | Bending stiffness of a cantilever

Find the equivalent spring constant of a cantilever beam of length l with Young's modulus E and area moment of inertia I, subjected to a concentrated end load F.



Tip-loaded cantilever; the illustrated load is labelled $F = W$.

Solution 16

Assume a uniform slender beam, linear elasticity and small deflection. Let s measure distance from the fixed end and let downward deflection $y(s)$ be positive.

1. Write the bending relation

The end force produces bending moment magnitude $F(l - s)$. With the chosen deflection sign, the beam equation is

$$EI\, y''(s) = F(l - s).$$

2. Integrate and apply the fixed-end conditions

Integrating once and using $y'(0) = 0$ gives

$$EI\, y'(s) = F\left(ls - \frac{s^2}{2}\right).$$

Integrating again and using $y(0) = 0$ gives

$$EI\, y(s) = F\left(\frac{ls^2}{2} - \frac{s^3}{6}\right).$$

3. Evaluate the tip deflection

At $s = l$,

$$\delta = y(l) = \frac{F}{EI}\left(\frac{l^3}{2} - \frac{l^3}{6}\right) = \frac{Fl^3}{3EI}.$$

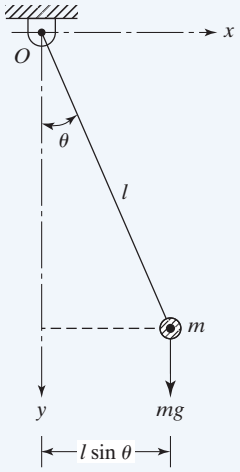
4. Identify the equivalent stiffness

$$\boxed{k = \frac{F}{\delta} = \frac{3EI}{l^3}}, \qquad \frac{(\mathrm{N/m^2})(\mathrm{m^4})}{\mathrm{m^3}} = \mathrm{N/m}.$$

Here I is an *area* second moment, in $\mathrm{m^4}$, not a mass moment of inertia. The stiffness relates transverse tip force to tip displacement.

Question 17 | Gravity as a restoring effect

Figure shows a simple pendulum of length l with a bob of mass m. For a small angular displacement θ, determine the *equivalent torsional spring constant* associated with the restoring moment due to gravity.



Simple pendulum.

Solution 17

Take θ positive away from the stable downward vertical position. Use radians in the small-angle approximation.

1. Find the moment arm of the weight

The weight mg acts vertically downward. Its perpendicular distance from the pivot is $l\sin\theta$.

2. Write the restoring moment with its sign

Gravity acts toward the equilibrium position, opposite the positive displacement:

$$M_g = -mgl\sin\theta.$$

3. Linearise for small angular motion

For $|\theta| \ll 1$ rad, $\sin\theta \approx \theta$, so

$$M_g \approx -mgl\theta.$$

4. Compare with a torsional spring

A linear torsional spring produces $M_s = -k_t\theta$. Therefore,

$$\boxed{k_t = mgl} \qquad [\mathrm{N\,m/rad}].$$

Energy check. The bob rises by $l(1-\cos\theta)$, so

$$U = mgl(1-\cos\theta) \approx \tfrac{1}{2}mgl\theta^2 = \tfrac{1}{2}k_t\theta^2.$$

The same stiffness follows. There is no physical spring here: gravity supplies an equivalent restoring effect near equilibrium.

Question 18 | Linearising a rubber mount

A precision milling machine, weighing 5000 N, is supported on a rubber mount. The force–deflection relationship of the rubber mount is given by

$$F = 375x + 0.1x^3$$

where F is in newtons and x in millimeters. Determine the equivalent linearized spring constant of the rubber mount at its static equilibrium position.

Solution 18

1. Find the static equilibrium

At equilibrium the mount force balances the machine weight:

$$375x_0 + 0.1x_0^3 = 5000.$$

The force law uses x in mm. Keep mm throughout this calculation.

2. Solve the equilibrium equation

Define $g(x) = 375x + 0.1x^3 - 5000$. Its derivative is $g'(x) = 375 + 0.3x^2 > 0$, so there is one real root. Newton's method gives

$$x_{n+1} = x_n - \frac{375x_n + 0.1x_n^3 - 5000}{375 + 0.3x_n^2}.$$

Starting from the linear estimate $x^{(0)} = 5000/375 = 13.3333$ mm:

Iteration	Displacement (mm)
0	13.333333
1	12.779939
2	12.777090
3	12.777090

Thus $\boxed{x_0 = 12.7771 \text{ mm}}$.

3. Evaluate the tangent stiffness

Differentiate the force law and evaluate at equilibrium:

$$k_t = \left.\frac{dF}{dx}\right|_{x_0} = 375 + 0.3(12.777090)^2 = \boxed{423.976 \text{ N/mm}}.$$

Since 1 N/mm = 1000 N/m,

$$\boxed{k_t = 4.23976 \times 10^5 \text{ N/m}}.$$

4. Interpret the linearisation correctly

For a small additional displacement $u = x - x_0$, the first-order expansion is

$$F(x_0 + u) \approx F(x_0) + k_t u = 5000 + 423.976u,$$

where u is in mm. Thus $\Delta F \approx k_t u$. The stiffness relates *changes* about equilibrium; $5000/k_t$ does not give x_0, because the tangent line does not pass through the origin.

Question 19 | Stiffness from measured force and length

One end of a helical spring is fixed and the other end is subjected to five different tensile forces. The lengths of the spring measured at various values of the tensile forces are given below:

Tensile force F (N)	0	100	250	330	480	570
Total length (mm)	150	163	183	194	214	226

Determine the force-deflection relation of the helical spring.

Solution 19

1. Distinguish total length from extension

The zero-load length is $L_0 = 150$ mm. Therefore $x = L - L_0$, giving

x (mm)	0	13	33	44	64	76
F (N)	0	100	250	330	480	570

2. Check whether a linear relation is reasonable

For the nonzero measurements, F/x is approximately

$$7.69,\ 7.58,\ 7.50,\ 7.50,\ 7.50 \quad \text{N/mm}.$$

The ratios are close, so a linear spring model is a reasonable fit over the measured range. The small differences are consistent with rounded measurements.

3. Fit a line through the measured zero-load point

Choose $F \approx kx$ and minimise the sum of squared force residuals:

$$S(k) = \sum_i (F_i - kx_i)^2.$$

Setting $dS/dk = -2\sum_i x_i(F_i - kx_i) = 0$ yields

$$k = \frac{\sum_i x_i F_i}{\sum_i x_i^2} = \frac{98110}{13066} = 7.5088 \text{ N/mm}.$$

Thus one suitable fitted relation is

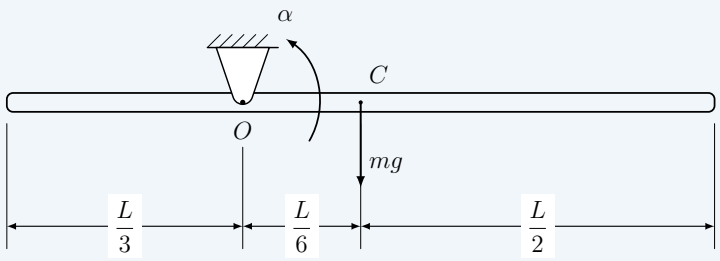
$$\boxed{F \approx 7.509x \quad (F \text{ in N},\ x \text{ in mm}).}$$

4. Interpret the result and convert units

To the precision suggested by the data, $F \approx 7.5x$ is also reasonable. In SI units the fitted stiffness is approximately 7509 N/m. If total length is used instead of extension, write $F \approx 7.509(L - 150)$ with lengths in mm, not $F = 7.509L$.

Question 20 | Mass moment of inertia and angular acceleration

A uniform slender bar of length L and mass m is hinged at O a distance $OC = \frac{L}{6}$ from its mass center C (see figure). The bar is released from rest in the *horizontal* position. Find the angular acceleration of the bar immediately after release.



Uniform bar and the forces at release.

Solution 20

Use the fixed pivot O as the moment reference. The bar's centre of mass is C, and the distance from O to C is $a = L/6$.

1. Find the mass moment of inertia about the centre

For a uniform slender bar, $dm = (m/L)\,ds$. Integrating along the bar gives

$$J_C = \int_{-L/2}^{L/2} s^2 \frac{m}{L}\,ds = \frac{m}{L}\left[\frac{s^3}{3}\right]_{-L/2}^{L/2} = \frac{mL^2}{12}.$$

2. Move the inertia reference to the pivot

The parallel-axis theorem gives

$$J_O = J_C + ma^2 = \frac{mL^2}{12} + m\left(\frac{L}{6}\right)^2 = \frac{mL^2}{9}.$$

3. Calculate the moment at release

At the horizontal position, the weight has perpendicular lever arm $L/6$. The hinge forces act through O and have zero moment about O. Thus

$$|M_O| = mg\frac{L}{6}.$$

4. Apply the fixed-axis moment equation

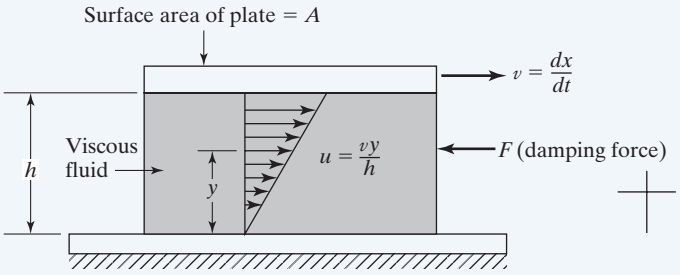
$$|M_O| = J_O|\alpha_0| \quad \Longrightarrow \quad \boxed{|\alpha_0| = \frac{mgL/6}{mL^2/9} = \frac{3g}{2L}}.$$

The rotation is in the direction that lowers the centre of mass. Its unit is s^{-2}, conventionally written $\mathrm{rad/s^2}$.

Although the bar is released from rest, its initial acceleration is not zero. For rotation about this fixed pivot, the total kinetic energy is $T = \frac{1}{2}J_O\dot{\theta}^2$; do not add a separate translation term to that expression. Mass is measured in kg, whereas weight mg is a force in N.

Question 21 | Viscous damping between plates

Two large parallel plates are separated by a uniform gap h filled with a Newtonian fluid of viscosity μ. One plate translates in its own plane at speed v relative to the other (see sketch). Derive the damping constant c relating the resisting force F to the plate speed, $F = c\,v$.



Parallel plates separated by a viscous fluid.

Solution 21

Let A be the wetted plate area. Assume no slip at either plate, a uniform gap, negligible edge effects and no pressure-driven flow. Let y measure distance from the stationary plate toward the moving plate.

1. Write the fluid velocity profile

No slip gives $u(0) = 0$ and $u(h) = v$. For the assumed linear profile,

$$u(y) = \frac{v}{h}y, \qquad \frac{du}{dy} = \frac{v}{h}.$$

2. Use Newton's law of viscosity

The shear stress magnitude is

$$\tau = \mu\frac{du}{dy} = \frac{\mu v}{h}.$$

3. Convert stress into force

Shear stress acts over area A, giving resisting force magnitude

$$F = \tau A = \frac{\mu A}{h}v.$$

4. Identify and check the damping constant

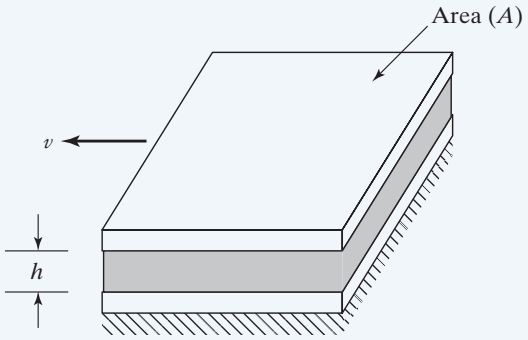
Comparing with $F = cv$,

$$\boxed{c = \frac{\mu A}{h}}, \qquad \frac{(\mathrm{N\,s/m^2})(\mathrm{m^2})}{\mathrm{m}} = \mathrm{N\,s/m}.$$

For signed velocities of two moving plates, the force on plate 1 is $F_1 = -c(v_1 - v_2)$. The force opposes *relative* motion. A larger gap reduces the velocity gradient and therefore reduces the damping coefficient.

Question 22 | Clearance of a lubricated bearing

A bearing can be approximated as two flat plates separated by a thin lubricant film of uniform thickness h. When SAE 30 oil (dynamic viscosity $\mu = 0.3445$ Pa $\cdot$ s) is used, the measured resisting force is $F = 400$ N at a relative plate speed $v = 10$ m/s. If the plate area is $A = 0.1$ m^2, determine the clearance h.



Flat-plate approximation of a lubricant film.

Solution 22

1. List the given quantities in SI units

$$\mu = 0.3445 \text{ Pa s} = 0.3445 \text{ N s/m}^2,$$
$$F = 400 \text{ N}, \qquad v = 10 \text{ m/s}, \qquad A = 0.1 \text{ m}^2.$$

2. Infer the damping coefficient

The measured force and speed give

$$c = \frac{F}{v} = \frac{400}{10} = 40 \text{ N s/m}.$$

3. Use the plate-film model to find the gap

From Question 21, $c = \mu A/h$. Rearranging,

$$h = \frac{\mu A}{c} = \frac{(0.3445)(0.1)}{40} = 0.00086125 \text{ m}.$$

4. Convert to millimetres and check

$$\boxed{h = 0.86125 \text{ mm} \approx 0.861 \text{ mm}}.$$

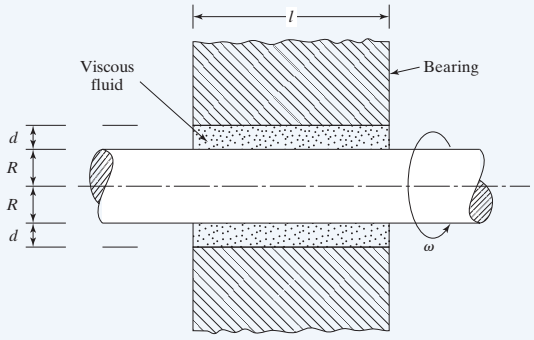
Substitution into the force law returns

$$F = \frac{\mu A v}{h} = \frac{(0.3445)(0.1)(10)}{0.00086125} = 400 \text{ N}.$$

The result is based on the given viscosity and the uniform, linear-velocity-profile model. The unit of $\mu A/c$ is metres, as required for a clearance.

Question 23 | Rotational damping of a journal bearing

A journal bearing supports a rotating shaft of radius R and length l. The radial clearance (film thickness) between shaft and bearing is d. The lubricant has viscosity μ and the shaft spins at angular speed ω. Assuming negligible leakage and a linear velocity profile across the film (concentric journal), derive the *rotational* damping constant c_t of the bearing such that $T = c_t\,\omega$.



Rotating shaft and surrounding lubricant film.

Solution 23

Use the stated concentric, linear-profile approximation. The thin-film model treats the gap as locally flat ($d \ll R$), with no slip at the shaft and stationary bearing surfaces.

1. Convert angular speed into surface speed

The shaft surface moves at $u_s = R\omega$. The bearing surface is stationary, so the speed difference across the film is $R\omega$.

2. Find the shear stress

For a linear velocity profile, the gradient magnitude is $R\omega/d$. Newton's viscosity law gives

$$\tau = \mu\frac{R\omega}{d}.$$

3. Calculate the tangential force magnitude

The wetted shaft area is circumference times length, $A = 2\pi Rl$. Thus

$$F = \tau A = \left(\frac{\mu R\omega}{d}\right)(2\pi Rl).$$

4. Convert the distributed shear into torque

Every shear contribution acts at radius R, so its torque adds about the shaft axis:

$$T = FR = \frac{2\pi\mu R^3 l}{d}\,\omega.$$

Comparing torque magnitudes with $T = c_t\omega$,

$$\boxed{c_t = \frac{2\pi\mu R^3 l}{d}} \qquad [\mathrm{N\,m\,s/rad}].$$

The signed resisting torque is $M_d = -c_t\omega$. This rotational coefficient relates torque to angular speed; its units differ from the translational coefficient c in Questions 21–22.

Question 24 | Gravity and the equilibrium coordinate

Indicate whether the following statement is true or false:
When a mass vibrates in a vertical direction, its weight can always be ignored in deriving the equation of motion.

Solution 24

False. Weight must be included in the force balance. For a linear spring, its explicit term cancels after the displacement is measured from static equilibrium.

1. Define displacement from the unstretched position

Consider the vertical mass and linear spring. Let y be positive downward from the position at which the spring is unstretched. Weight is $+mg$ and spring force is $-ky$. Newton's law gives

$$m\ddot{y} = mg - ky.$$

2. Find the static sag

At rest, $\ddot{y} = 0$. Hence

$$k\delta_{\text{st}} = mg, \qquad \delta_{\text{st}} = \frac{mg}{k}.$$

3. Measure motion from static equilibrium

Set $y = \delta_{\text{st}} + x$. Since δ_{st} is constant, $\ddot{y} = \ddot{x}$. Substituting gives

$$m\ddot{x} = mg - k(\delta_{\text{st}} + x) = \underbrace{mg - k\delta_{\text{st}}}_{0} - kx,$$

so $\boxed{m\ddot{x} + kx = 0}$.

4. Include the damper and external force used in class

A parallel viscous damper connected to the fixed base contributes $-c\dot{y}$ and an applied downward force contributes $F(t)$. The same shift gives

$$m\ddot{y} = mg + F(t) - c\dot{y} - ky \quad \Longrightarrow \quad \boxed{m\ddot{x} + c\dot{x} + kx = F(t).}$$

All terms in the boxed equation have units of force. At release with $F = 0$ and $\dot{x} = 0$, the acceleration is $\ddot{x} = -kx/m$, even though the velocity is zero.

5. State the limitation

Gravity still determines the static deformation. With a nonlinear spring, it can also change the tangent stiffness through the equilibrium position. In a pendulum it supplies the restoring moment itself. Weight therefore cannot *always* be ignored.

Question 25 | Initial conditions and a complete response

How many arbitrary constants must a general solution to a second-order differential equation have? How are these constants determined?

Solution 25

1. Count the constants

For the regular second-order motion equation considered here, the general solution contains **two independent arbitrary constants**. Integrating a second-order equation requires two pieces of information about the motion.

2. Specify the initial conditions

For a vibration initial-value problem, the usual data are

$$x(0) = x_0, \qquad \dot{x}(0) = v_0.$$

These specify the initial displacement and initial velocity. Position alone does not tell us whether the mass is moving, or how fast.

3. Determine the constants

After finding the general response, substitute $t = 0$ into the response and its first derivative. This produces two equations for the two unknown constants. Solve those equations and substitute the constants back into the response.

4. Distinguish required data from derived quantities

For $m\ddot{x} + c\dot{x} + kx = F(t)$ with $m > 0$, the initial acceleration follows from the equation:

$$\boxed{\ddot{x}(0) = \frac{F(0) - cv_0 - kx_0}{m}.}$$

It is not a third independent initial condition. The model parameters, excitation history and two initial conditions together define the initial-value problem. No explicit solution of the differential equation is required at this stage.